

Semester: August 2021- November 2021 Examination: In Semester Examination		
Programme code: Programme : B.Tech. Computer Engineering	Class: SY	Semester: III (SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering	Name of the department/Section/Center: COMP	
Course Code: 116U01C305	Name of the Course: Discrete Mathematics (Specially Abled)	

Question No.		Max. Marks
Q1	Multiple Choice Questions: 1. The proposition $\sim p \vee (p \vee q)$ is a A) Tautology B) Contradiction C) Logical equivalence D) None of the above 2. Translate $\forall x \exists y (x < y)$ in English, considering domain as a real number for both the variable. A) For all real number x there exists a real number y such that x is less than y B) For every real number y there exists a real number x such that x is less than y C) For some real number x there exists a real number y such that x is less than y D) For each and every real number x and y such that x is less than y 3. The binary relation $\{(1,1), (2,1), (2,2), (2,3), (2,4), (3,1), (3,2)\}$ on the set $\{1, 2, 3\}$ is _____ a) reflexive, symmetric and transitive b) irreflexive, symmetric and transitive c) neither reflexive, nor irreflexive but transitive d) irreflexive and antisymmetric 4. The Cartesian Product $B \times A$ is equal to the Cartesian product $A \times B$. A) True B) False 5. The members of the set $S = \{x \mid x \text{ is the square of an integer and } x < 100\}$ is _____ A) $\{0, 2, 4, 5, 9, 58, 49, 56, 99, 12\}$ B) $\{0, 1, 4, 9, 16, 25, 36, 49, 64, 81\}$ C) $\{1, 4, 9, 16, 25, 36, 64, 81, 85, 99\}$ D) $\{0, 1, 4, 9, 16, 25, 36, 49, 64, 121\}$	10 marks

	6. The binary relation $\{(1,1), (2,1), (2,2), (2,3), (2,4), (3,1), (3,2)\}$ on the set $\{1, 2, 3\}$ is _____ A) reflexive, symmetric and transitive B) irreflexive, symmetric and transitive C) neither reflexive, nor irreflexive but transitive D) irreflexive and antisymmetric 7. Consider the relation: $R' (x, y)$ if and only if $x, y > 0$ over the set of non-zero rational numbers, then R' is _____ A) not equivalence relation B) an equivalence relation C) transitive and asymmetry relation D) reflexive and antisymmetric relation 8. Negation of statement $(A \wedge B) \rightarrow (B \wedge C)$ is _____ A) $(A \wedge B) \rightarrow (\sim B \wedge \sim C)$ B) $\sim(A \wedge B) \vee (B \vee C)$ C) $\sim(A \rightarrow B) \rightarrow (\sim B \wedge C)$ D) None of the mentioned 9. Let R_1 be a relation from $A = \{1, 3, 5, 7\}$ to $B = \{2, 4, 6, 8\}$ and R_2 be another relation from B to $C = \{1, 2, 3, 4\}$ as defined below: i. An element a in A is related to an element b in B (under R_1) if $a * b$ is divisible by 3. ii. An element a in B is related to an element b in C (under R_2) if $a * b$ is even but not divisible by 3. Which is the composite relation $R_1 R_2$ from A to C? A) $R_1 R_2 = \{(1, 2), (1, 4), (3, 3), (5, 4), (5, 6), (7, 3)\}$ B) Φ C) $R_1 R_2 = \{(1, 2), (1, 6), (3, 2), (3, 4), (5, 4), (7, 2)\}$ D) $R_1 R_2 = \{(2, 2), (3, 2), (3, 4), (5, 1), (5, 3), (7, 1)\}$ 10. If set A and B have 3 and 4 elements respectively then the number of subsets of set $(A \times B)$ is? A) 1024 B) 2048 C) 512 D) 4096	
Q.2	Solve <u>ANY TWO</u> of the following:	10 marks (or 5+5)
(i)	Prove Following Implications using Laws of Logic: a) $\sim(p \leftrightarrow q) \equiv \sim((p \rightarrow q) \wedge (q \rightarrow p))$ b) $((p \vee \sim q) \wedge (\sim p \vee \sim q)) \vee q$ is a tautology	(3 M) (2 M)
(ii)	Write English Sentences corresponding to following:	

(iii)	a) $\forall x \exists y R(x, y)$ b) $\exists x \forall y R(x, y)$ c) $\forall x (\sim Q(x))$ d) $\exists y (\sim P(y))$ e) $\forall (P(x))$ Where, $P(x) : x \text{ is even}$ $Q(x) : x \text{ is prime no.}$ $R(x, y) : x + y \text{ is even}$ Show that $\frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n} > \frac{13}{24} \quad \forall n \geq 2$ by mathematical induction	
Q 3	Solve <u>ANY TWO</u> of the following: (i) Let $A = \{1, 2, 3, 4, 5\}$ $R = \{(1,2),(1,3),(1,4),(1,5),(2,3), (2,4), (2,5), (3,4), (3,5),(4,5)\}$ i. Draw diagram of R ii. Compute: R^2 iii. Draw diagram of R^2 (ii) Let $A = \{1, 2, 3, 4\}$ $R = \{(1,1),(1,2),(2,3),(2,4),(3,4),(4,1),(4,2)\}$ $S = \{(3,1),(4,4),(2,3),(2,4),(1,1),(1,4)\}$ i. Compute: $R \circ R$ ii. Compute: $S \circ S$ (iii) Let $A = \{1, 2, 3, 4\}$ $R = \{(1,1),(1,2),(1,4),(2,4),(3,1),(3,2),(4,2), (4,3), (4,4)\}$ i. What is transitive closure? ii. Obtain M_R i.e. matrix representation of relation set R.	5+5

	iii. By using 'Warshall's Algorithm' obtain W_0 , W_1	